

APP-DEV 2 PROJECT REPORT

1. Student Details

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About Me

I am a student in the IIT Madras BS Degree Program (Diploma Level) with an interest in full-stack web development. I enjoy building practical web applications and learning how different technologies work together. This project helped me gain experience with Flask, Vue.js, REST APIs, Redis caching, Celery background tasks, and modern web application development.

2. Project Details

Project Title: Placement Portal Application

Problem Statement

The objective of this project is to build a modern placement portal that connects students, companies, and administrators through a role-based web application. The portal automates the placement process including company registration, job postings, student applications, approvals, notifications, and placement management.

Approach

The application is developed using Flask as the backend REST API and Vue.js as the frontend framework. JWT authentication is used for secure login and role-based authorization. Redis caching is implemented to improve API performance, while Celery handles background tasks such as email notifications and CSV export. SQLite is used as the database with SQLAlchemy ORM.

3. AI/LLM Declaration

AI tool ChatGPT was used in a limited capacity (approximately 15-20%) during the development of this project. The assistance was mainly used for debugging errors, understanding concepts, UI improvements, and learning new technologies such as Redis, Celery, and Vue.js. All core logic, database design, API implementation, and application development were completed manually.

4. Technologies and Frameworks Used

- Flask – Backend REST API development
 - Vue.js 3 – Frontend framework
 - Vue Router – Client-side routing
 - Axios – API communication
 - SQLAlchemy – ORM for database operations
 - SQLite – Database
 - Flask JWT Extended – Authentication using JWT
 - Redis – API caching
 - Flask-Caching – Redis integration
 - Celery – Background task processing
 - Flask-Mail – Email notifications
 - Bootstrap 5 – User Interface
 - Vite – Vue development server and build tool
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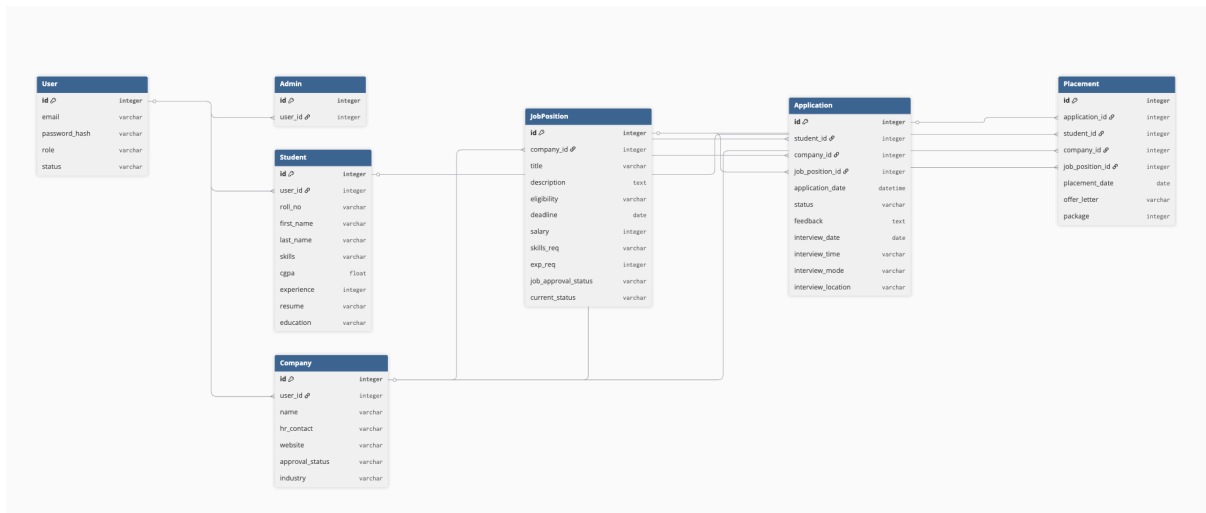
5. Database Schema

Tables

1. User
2. Admin
3. Student
4. Company
5. JobPosition
6. Application
7. Placement

Relationships

- One-to-One → User → Student
- One-to-One → User → Company
- One-to-One → User → Admin
- One-to-Many → Company → JobPosition
- One-to-Many → Student → Application
- One-to-Many → JobPosition → Application
- One-to-Many → Student → Placement
- One-to-Many → JobPosition → Placement
- One-to-One → Application → Placement



6. Architecture and Features

Architecture

- **app.py** – REST API routes, authentication, business logic
- **models.py** – SQLAlchemy database models
- **frontend/** – Vue.js frontend application
- **components/** – Reusable Vue components
- **views/** – Individual dashboard pages
- **router/** – Vue Router configuration
- **celery_worker.py** – Celery background worker
- **Redis** – API caching
- **SQLite** – Database storage

Implemented Features

- Role-based authentication (Admin, Student, Company)
- JWT-based login system
- Student registration
- Company registration and approval
- Admin dashboard
- Student dashboard
- Company dashboard
- Job creation and management
- Student job application system
- Placement management
- Search and filtering of jobs
- CSV export of application history
- Email notification support using Celery
- Redis API caching
- Automatic cache invalidation
- Progressive Web App (PWA) support

- Responsive mobile-friendly interface
 - Password hashing for security
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7. Additional Features Implemented

- Redis cache integration for frequently used APIs
 - Celery background task processing
 - CSV export functionality
 - Progressive Web App (PWA)
 - Mobile responsive interface
 - Search and filtering functionality
 - Automatic cache refresh after database updates
 - JWT authentication with protected routes
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8. Video Presentation URL

 *APP DEV 2 PROJ VIDEO.MP4*